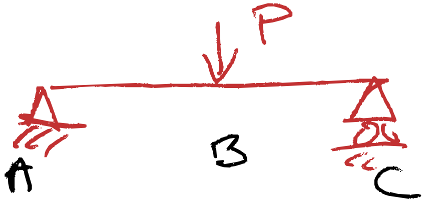
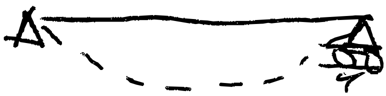
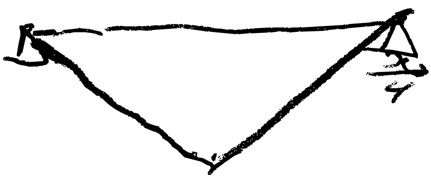
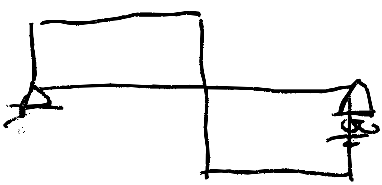


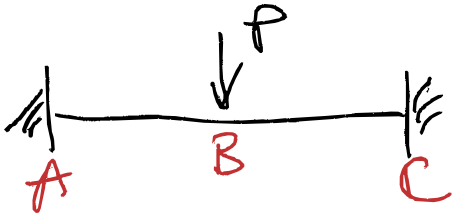


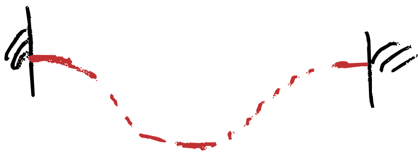
BMD



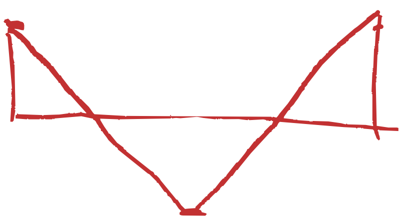
SFD



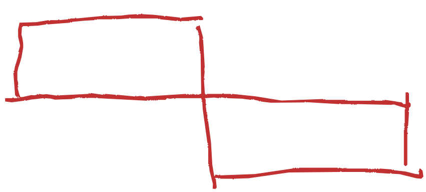


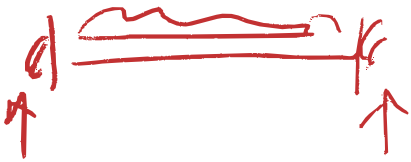


BMD

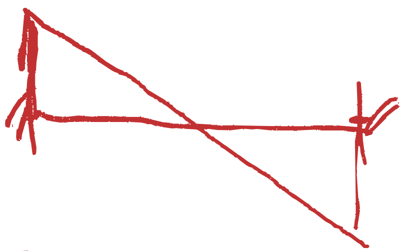


SFD



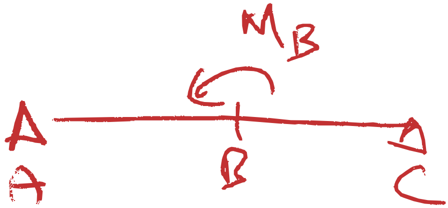


SFD

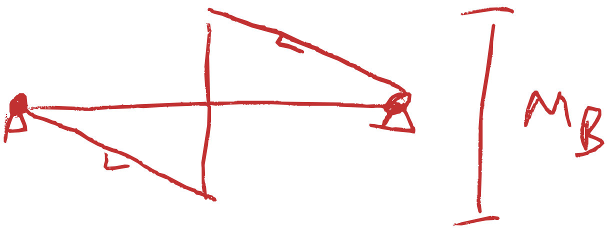


BMD

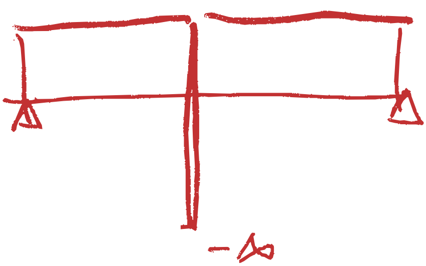
x quadratic

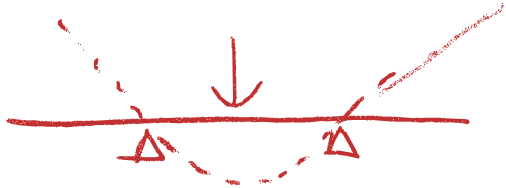


BMD

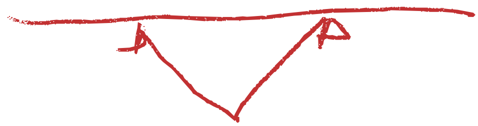


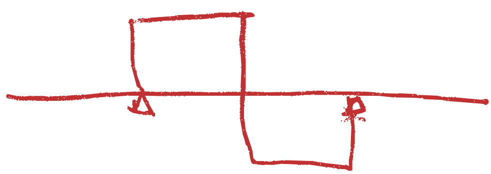
SFD

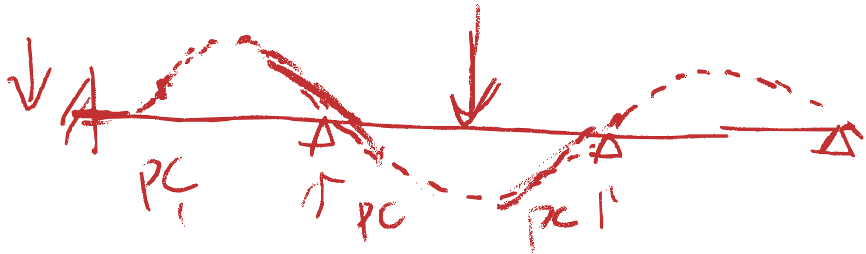




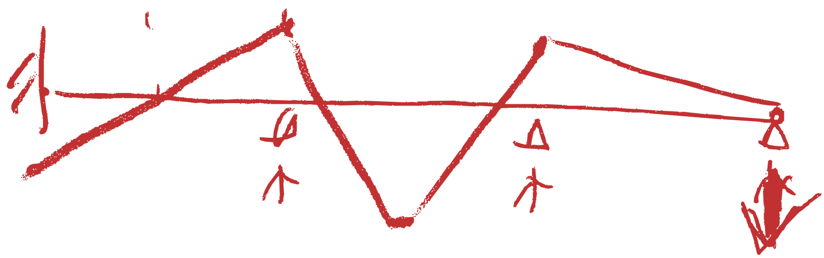
BMD



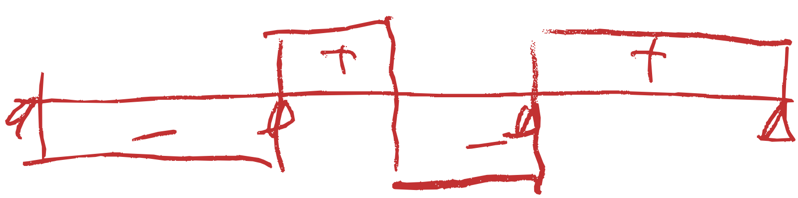


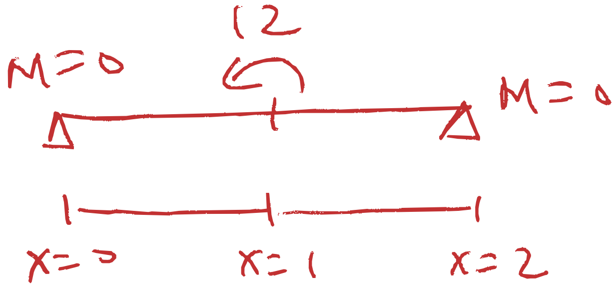


BMD



SFD





$$w(x) = -12\langle x-1\rangle^{-2} = \frac{dV}{dx}$$

$$V(x) = -12\langle x-1\rangle^{-1} + C_1$$

$$M(x) = -12\langle x-1\rangle^{0} + C_1 x + C_2$$

$$M(0) = -12\langle 0-1\rangle^{0} + C_1 \cdot 0 + C_2$$

$$= 0 \Rightarrow C_2 = 0$$

$$M(2) = -12\langle 2-1\rangle^{0} + C_1 \cdot 2$$

$$= -12 + 2C_1 = 0$$

$$\Rightarrow C_1 = 6$$

$$M(x) = -12\langle x-1\rangle^{0} + 6x$$

$$M(x) = -12\langle x-1\rangle^0 + 6x$$

$$\kappa(x) = \frac{M(x)}{EI} = M(x)$$

$$= -12\langle x-1\rangle^0 + 6x$$

$$\theta(x) = -12\langle x-1\rangle^1 + 3x^2 + C_3$$

$$v(x) = -\frac{12}{1+1}\langle x-1\rangle^2 + x^3 + C_3 x$$

$$+ C_4$$

$$= -6\langle x-1\rangle^2 + x^3 + C_3 x + C_4$$

$$v(0) = -6\langle 0-1\rangle^2 + 0^3 + C_3 \cdot 0 + C_4$$

$$= 0 \Rightarrow C_4 = 0$$

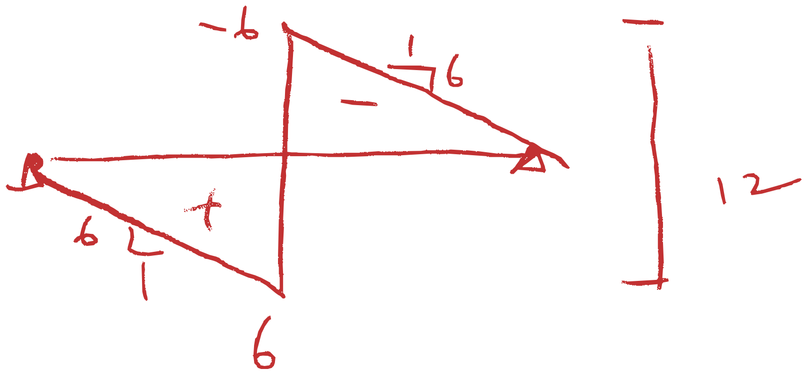
$$v(2) = -6\langle 2-1\rangle^2 + 2^3 + 2 \cdot C_3$$

$$= -6 + 8 + 2C_3 = 0$$

$$\Rightarrow C_3 = -1$$

$$v(x) = -6\langle x-1\rangle^2 + x^3 - x$$

$$M(x) = -12\langle x-1\rangle^{0} + 6x$$



$$V(x) = -12\langle x-1\rangle^{-1} + 6$$

SFD